

Roll: 2103064

Task-0:

Code:

```
mkdir lab-test
cd lab-test
mkdir task-1
mkdir task-2
ls
```

Output:

```
adil@ASUS-Vivobook-Intel-corei7:~/lab-test$ mkdir task-1
adil@ASUS-Vivobook-Intel-corei7:~/lab-test$ mkdir task-2
adil@ASUS-Vivobook-Intel-corei7:~/lab-test$ ls
task-1  task-2
```

Task-1:

Code:

```
#!/bin/bash

if [ $1 -lt 0 ]; then
echo "Error occurred as argument is negative"
elif [ $2 -lt 0 ]; then
echo "Error occurred as argument is negative"
elif [ $3 -lt 0 ]; then
echo "Error occurred as argument is negative"
elif [ $4 -lt 0 ]; then
echo "Error occurred as argument is negative"
else
sum=$(( $1+$2+$3+$4 ))
touch file$sum.txt
fi
```

Output:

```
adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ ./create_script.sh 1 2 3 4
./create_script.sh: line 3: 0: No such file or directory
./create_script.sh: line 5: 0: No such file or directory
./create_script.sh: line 7: 0: No such file or directory
./create_script.sh: line 9: 0: No such file or directory
adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ ls
create_script.sh  file10.txt
adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ █

adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ ./create_script.sh 1 2 3 -4
Error occurred as argument is negative
adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ ./create_script.sh 1 -2 3 4
Error occurred as argument is negative
adil@ASUS-Vivobook-Intel-corei7:~/lab-test/task-1$ █
```

Task-2:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <string.h>
```

```

#include <math.h>

struct new_struct {
    int start;
    int end;
};

void* squire_root(void* ptr)
{
    struct new_struct* arg=(struct new_struct*)ptr;
    int len=arg->end-arg->start+1;
    double *res=(double*)malloc(sizeof(double)*len);

    int k=0;
    for(int i=arg->start;i<=arg->end;i++)
    {
        res[k]=sqrt(i);
        k++;
    }

    return (void*)res;
}

int main(int arvc, char *arvs[])
{
    pthread_t t1,t2;
    struct new_struct s1, s2;

    int st,end;
    if(arvc>1)
    {
        st=atoi(arvs[1]);
        end=atoi(arvs[2]);
    }

    int mid=(st+end)/2;

    s1.start=st;
    s1.end=mid;

    s2.start=mid+1;
    s2.end=end;
    printf("Threads are Generating and doing Calculation\n");

    pthread_create(&t1,NULL,squire_root,&s1);
    pthread_create(&t2,NULL,squire_root,&s2);

    double *res1,*res2;

    pthread_join(t1,(void**)&res1);
    pthread_join(t2,(void**)&res2);

    printf("Threads1:\n");
    for(int i=0;i<mid;i++)
    {
        printf("%lf\n",res1[i]);
    }
    printf("Threads1:\n");
}

```

```
    for(int i=0;i<mid;i++)
    {
        printf("%lf\n",res2[i]);
    }

    free(res1);
    free(res2);

    return 0;
}
```

Output: